

6,906: Economics: Macro for Computer Science

LECTURE 8: INFLATION, UNEMPLOYMENT, AND THE PHILLIPS CURVE

Maria Bolboaca

University of St. Gallen

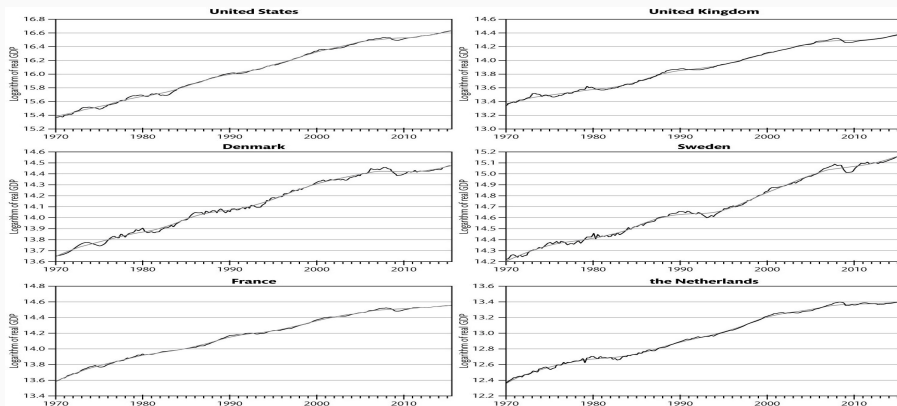
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Learning objectives

- By the end of this lecture, you should be able to:
 - Discuss the differences between the short- and the medium-run analysis of the economy;
 - Use a simple economic model to analyze the labor market;
 - Study how the relation between inflation and unemployment depends on inflation expectations.

GDP over time

So far, we studied what drives long-term growth (the trend) - Part 2, analyzed short-term fluctuations - Part 3, and now focus on explaining the values on the trend line at any given period - Part 4.



Source: Sorensen and Whitta-Jacobsen (2022)

Moving from the short- to the medium-run analysis of the economy

- In the short-run analysis of the economy, we assumed that firms could supply any amount of output at a given price level.
- The IS-LM model implies that variations in the demand for goods and services, driven by consumer spending or monetary and fiscal policy, determine the level of output produced (demand $\rightarrow$ production).
- Ignoring prices (and resource scarcity) is fine when focusing on the short run, but not when studying the medium run.
- For example, when the government increases its spending, firms may try to meet the increased demand by asking the employees to work overtime in the short run. However, in the medium run, this overtime is reduced or paid out, leading to price and wage increases.
- Today, we explore how prices and wages change over time, and how this affects output in a model suited for the short- and medium-run analysis of the economy.

Model of labor market

Labor supply - Wage determination:

- Three key factors in wage negotiations: expected price level, bargaining power of the two sides, and reservation wages.
- This is captured in the wage determination equation:

$$W = P^e F(u, z)$$

(−, +)

- Nominal wages W depend positively on the expected price level P^e , because when nominal wages are set, the relevant price levels are still unknown (e.g., some contracts are set in advance for a couple of years). Note that workers and firms do not care about nominal wages, but real wages (W/P).
- Wages decrease with unemployment u since higher unemployment either weakens workers' bargaining power or allows firms to pay lower wages, while keeping workers willing to work.
- All other factors that affect wages positively are captured by variable z , e.g., unemployment benefits, employment protection, etc.

Model of labor market

Labor demand - Price determination:

- Prices depend on the firms' costs, which in turn depend on the nature of the production function.
- The production function is:

$$Y = AN$$

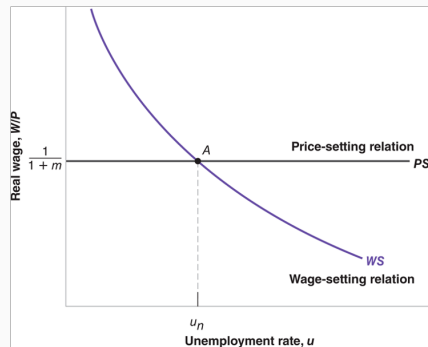
- Output is determined by labor augmenting productivity A and employment N .
- Marginal cost of an additional unit of labor is W/A .
- In perfect competition, $P = W/A$, but under the assumption that goods markets are not competitive: $P = (1 + m)W/A$, where m is the markup of price over the marginal cost.
- Simplification: $A = 1$.

Equilibrium of the labor market:

- If the actual W depends on the actual price level (P) rather than the expected price level (P^e), wage setting and price setting determine the equilibrium rate of unemployment, also called the natural or structural rate of unemployment, u_n .
- Wage setting (WS) relation: $\frac{W}{P} = F(u, z)$
- Price setting (PS) relation: $\frac{W}{P} = \frac{1}{(1+m)}$

Model of labor market

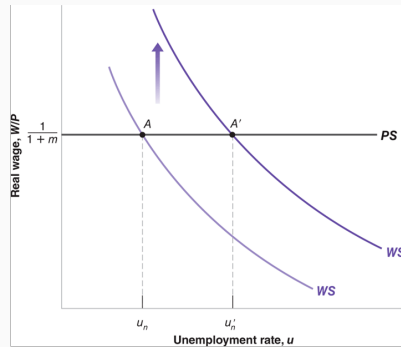
- Equilibrium of the labor market implies equating the wage-setting and price-setting relations.
- This leads to the medium-run equilibrium condition: $F(u, z) = \frac{1}{(1+m)}$, which implies that equilibrium unemployment depends on both m and z .



Source: Blanchard (2021)

Model of labor market

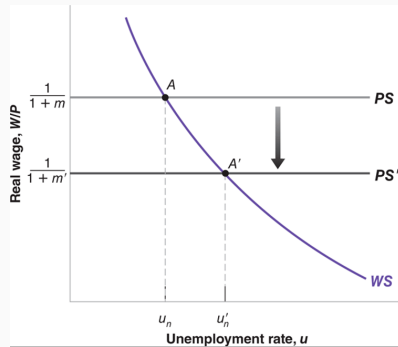
- An increase in unemployment benefits leads to an increase in the natural rate of unemployment.



Source: Blanchard (2021)

Model of labor market

- An increase in the markup leads to an increase in the natural rate of unemployment.



Source: Blanchard (2021)

From the natural rate of unemployment to the natural level of output

- The natural level of employment is given by $N_n = L(1 - u_n)$
- Hence, the natural level of output (also known as potential output)
$$Y_n = AN_n = AL(1 - u_n)$$
- Y_n corresponds to the points in time on the trend line of our real GDP graphs.
- Under the assumption that the expected price level equals the actual price level, equilibrium in the labor market determines the natural unemployment rate and the associated level of output.
- This assumption, i.e., $P^e = P$, holds for the medium run, as expectations are unlikely to be systematically wrong forever, but there is no reason to assume it true in the short run.
- In the short run, the factors studied before, e.g., monetary policy, fiscal policy, and consumer spending, determine the level of output, which may deviate from potential output.
- For example, an increase in government spending may take production above potential output, which makes wages (and thus goods prices) rise.

The Phillips curve

- Lecture 2: The Phillips curve is derived from the data and shows that a low unemployment rate is associated with an increase in the inflation rate.
- We now use our labor model to explain the Phillips curve, by allowing $P^e \neq P$ in the short run.
- We replace W in the price setting equation $P = (1 + m)W$ with the expression given by the wage setting relation: $W = P^e F(u, z)$ to get:

$$P = (1 + m)P^e F(u, z)$$

- Let's assume $F(u, z) = 1 - \alpha u + z$, to rewrite the expression above as $P = (1 + m)P^e(1 - \alpha u + z)$.
- This expression can be further transformed in terms of the inflation rate π and expected inflation rate π^e as:

$$\pi = \pi^e + (m + z) - \alpha u$$

The Phillips curve

- The Phillips curve relation, written with a time subscript, is

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t,$$

where $\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}$ and $\pi_t^e = \frac{P_t^e - P_{t-1}}{P_{t-1}}$

- The equation describes the Phillips curve, as it implies that, for given values of π^e , m , and z , a decrease in the unemployment rate u leads to an increase in the inflation rate π .
- Also, an increase in π^e leads to an increase in π ; while, given π^e , an increase in m or z leads to an increase in π .
- How are inflation expectations formed?

Anchored inflation expectations and the original Phillips curve

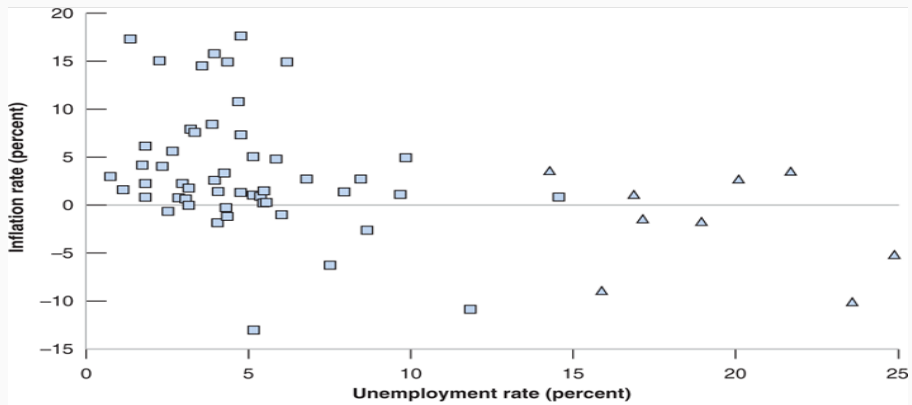
- If economic agents expect the inflation rate to vary around a given level $\bar{\pi}$, then we say they have anchored expectations, i.e., $\pi_t^e = \bar{\pi}$, and the Phillips curve relation becomes

$$\pi_t = \bar{\pi} + (m + z) - \alpha u_t.$$

- In this case, we observe the negative relation between unemployment and inflation discovered by Phillips, Samuelson, and Solow.

The Phillips curve and its mutations

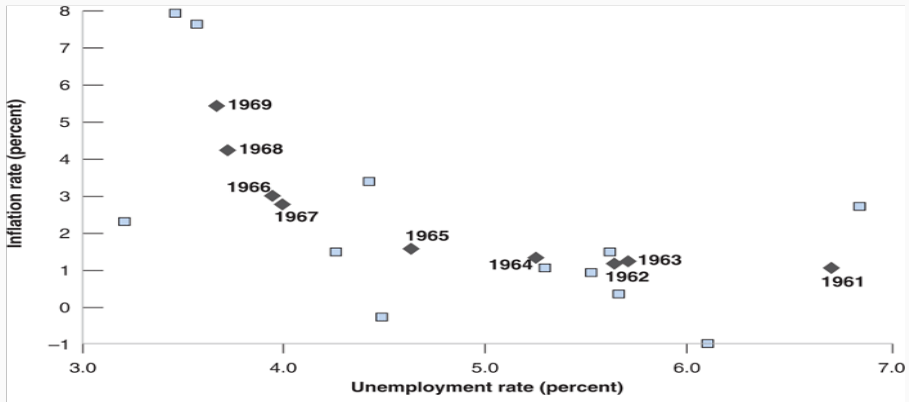
During 1900-1960 in the United States, a low unemployment rate was associated with a high inflation rate, and a high unemployment rate was associated with a low or negative inflation rate.



Source: Blanchard (2021)

The Phillips curve and its mutations

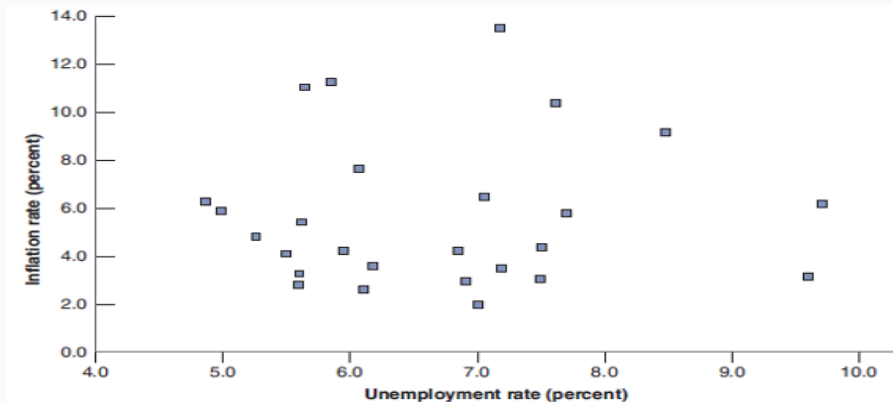
The steady decline in the U.S. unemployment rate throughout the 1960s was associated with a steady increase in the inflation rate, confirming the Phillips curve predictions.



Source: Blanchard (2021)

The Phillips curve and its mutations

Data for 1970-1995 for the United States indicate that the correlation between unemployment and inflation disappeared.



Source: Blanchard (2021)

De-anchored inflation expectations and the accelerationist Phillips curve

- Why did the original Phillips curve vanish in the U.S. at the beginning of the 1970s?
- Because economic agents changed the way they formed their inflation expectations. This change was because the rate of inflation became more persistent.
- Suppose economic agents form inflation expectations according to:

$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1}$$

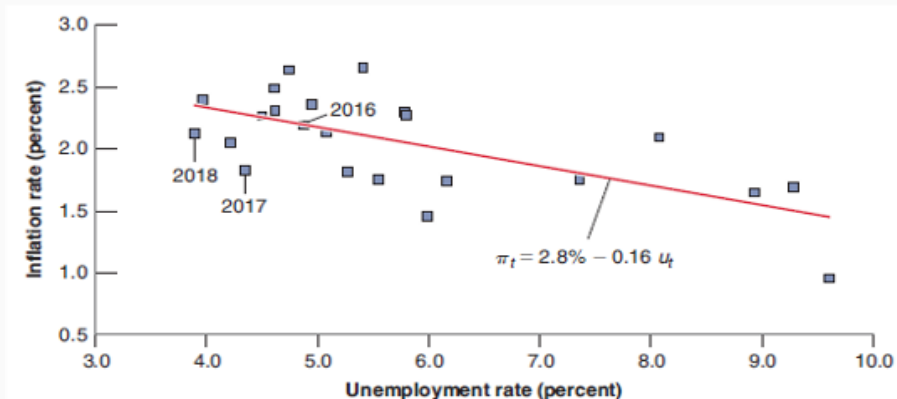
- Anchored expectations $\rightarrow \theta = 0$
- De-anchored expectations $\rightarrow \theta = 1$
- Hence, due to persistent inflation increases, θ rose to 1, so that the Phillips curve equation changes to:

$$\pi_t - \pi_{t-1} = (m + z) - \alpha u_t,$$

- This implies that the unemployment rate influences the change in the inflation rate $\rightarrow$ accelerationist Phillips curve

The Phillips curve and its mutations

Since the mid-1990s, the Phillips curve in the U.S. has again taken the form of a relation between the inflation rate and the unemployment rate.



Source: Blanchard (2021)

Re-anchoring of inflation expectations

- In the 1990s, the Phillips curve relationship changed once more, driven by changes in monetary policy.
- Beginning in the early 1980s, central banks-including the Federal Reserve-increasingly prioritized price stability, often targeting inflation rates around 2%.
- One approach - use of Taylor rule $i_t = r + \pi_t + \gamma(\pi_t - \bar{\pi}) + \sigma(Y_t - Y_{n,t})$, where parameters γ and σ determine how much value the central bank puts on price stability and economic growth, respectively.
- By the mid-1990s, the Fed had largely achieved this price stability objective. As a result, inflation expectations, which had become de-anchored during the 1970s and 1980s, were re-anchored, reflecting greater confidence in the central bank's commitment to low and stable inflation.

The Phillips curve and the natural rate of unemployment

- In the medium run, $\pi_t = \pi_t^e$, which in the Phillips curve relation implies that $u_n = \frac{m+z}{\alpha}$
- Hence, the Phillips curve relation can be rewritten as: $\pi_t - \pi_t^e = -\alpha(u_t - u_n)$.
- The Phillips curve is the relationship between the change in the inflation rate and the deviation of the unemployment rate from the natural unemployment rate.
- This is the reason why u_n is also known as the non-accelerating inflation rate of unemployment (NAIRU).

Summary

- Today we learned:
 - To determine the equilibrium unemployment rate and corresponding output level in a simple labor market model.
 - To analyze the relation between inflation and unemployment based on how economic agents form inflation expectations.
- The IS-LM model is suitable for analyzing the economy in the short run. Still, in the medium run, we must consider resource constraints (labor market - Y_n, u_n) and price changes (inflation).
- Therefore, we extend the IS-LM model with the Phillips curve (PC curve) in the next class and also learn how to introduce expectations in the model to study economic fluctuations and the effects of policy actions. .